Machine Learning Aided NR-V2X Quality of Service Predictions

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***Abstract*—Vehicle-to-Everything Communication (V2X) technologies aim to meet strict quality-of-service (QoS) requirements of vehicular connectivity applications such as safety message exchange, remote driving, and sensor data sharing. The high reliability requirement is particularly important to enable safety relevant applications. Thus, predicting QoS levels becomes key to ensure the reliability of the connected vehicle applications. Recently, machine learning (ML) algorithms are demonstrated to provide dependable predictions to plan, simulate, and evaluate the performance of vehicular networks. In this paper, we propose ML aided New Radio (NR)-V2X QoS predictions scheme to provide Packet Delivery Ratio (PDR) and throughput predictions with the input of Modulation and Coding Schemes (MCS), distance-to-base station, Signal to Interference plus Noise Ratio (SINR), and packet size. Seven different ML algorithms based prediction models are trained and evaluated by using NR-V2X simulation data. We provide performance comparisons between Support Vector Regression (SVR), Deep Neural Network (DNN), Random Forest (RF), Gradient Boosting Machine (GBM), Extreme Gradient Boosting (XGBoost), Categorical Boosting (CatBoost), and Light GBM (LGBM) for predicting throughput and PDR. We demonstrate that CatBoost and RF are the best performing algorithms to predict throughput and PDR of NR-V2X networks, respectively.**

I. Introduction

V2X technology provides connectivity between vehicles and other road users to enhance road safety, efficiency, and sustainability. Currently, IEEE 802.11p and C-V2X technologies are standardized to support numerous V2X applications with different QoS requirements in terms of reliability, data rate and latency [1] [2]. IEEE 802.11p is an evolved version of the IEEE 802.11 standard, which includes enhancements to support Intelligent Transportation Systems (ITS) and defines communication protocols for Dedicated Short-Range Communications (DSRC). The 3rd Generation Partnership Project (3GPP) standardizes Cellular Vehicle-to-Everything (C-V2X) under Release 14 based on cellular technology for V2X services. NR-V2X is a 3GPP standard and is expected to be particularly suitable for safety relevant applications. NR-V2X under Release 16 and 17 is anticipated to enable advanced V2X use cases that require high communication speed with ultra-high reliability and low latency such as remote driving, platooning, and extended sensor information exchange [3].

Prediction of V2X QoS is key to improve QoS and support automation where V2X network conditions need to be enhanced for fast and reliable communication. Moreover, V2X applications can benefit from the predictive QoS by optimizing application parameters such as data packet size, data transmission intervals, and cache durations. In the literature, QoS prediction for V2X is explored in [4] [5] [6] [7] [8] [9]. In [4], end-to-end (E2E) QoS prediction is studied utilizing ML algorithms, particularly supervised learning and time series algorithms. In [5], 3GPP standardized Network Data Analytics Function (NWDAF) under Release 16 is described to gather data over 5G networks and provide data analytics to improve QoS, Artificial Intelligence (AI) and ML algorithms. In [6], the performances of different ML algorithms are evaluated with input variables of only SINR, only location, SINR and location for teleoperation and local map sharing. DNN and RF are proven to ensure a decent uplink (UL) throughput prediction in that framework. [7] proposes a predictive QoS model based on fusion machine learning approaches for V2X communications beyond 5G systems, which handles issues from the point of spatial, temporal, and system modeling. The main goal of the presented study is to predict various QoS metrics such as data rate and E2E delay to assign resources in an effective manner. In [8], an LSTM-based time series model is provided, where location, speed, and distance of the teleoperated vehicle, cell load percentage in resource blocks (RB)s and number of vehicles connected in the cell are used as inputs, to predict UL throughput of a teleoperated vehicle. [9] presents a scheme based on supervised learning algorithms and ARIMA algorithm so as to predict C-V2X QoS levels with reference-signal-received-power (RSRP), SINR, Channel Quality Indicator (CQI), the User-Equipment (UE)'s historical throughput data. However, to date, machine learning models employed to forecast QoS levels of V2X networks mainly depend on UE's coordinates and QoS metrics such as SINR, RSRP, Reference Signal Received Quality (RSRQ) and throughput without considering configuration parameters of V2X networks such as MCS.

In this paper, we propose an ML based model for the prediction of PDR and throughput in NR-V2X with MCS, packet size, distance-to-base station and SINR inputs. To generate

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realistic data, we first create different simulation scenarios for NR-V2X through Simulation of Urban MObility (SUMO) simulator and 5G-Lena simulator to generate a dataset. Then, we perform offline training with ML models and evaluate the performance of these models in terms of PDR and throughput estimation accuracy.

The rest of the paper is organized as follows. Section II describes system model. Section III introduces proposed ML based prediction scheme. Section IV provides performance evaluations for the proposed model. The paper is concluded in Section V.

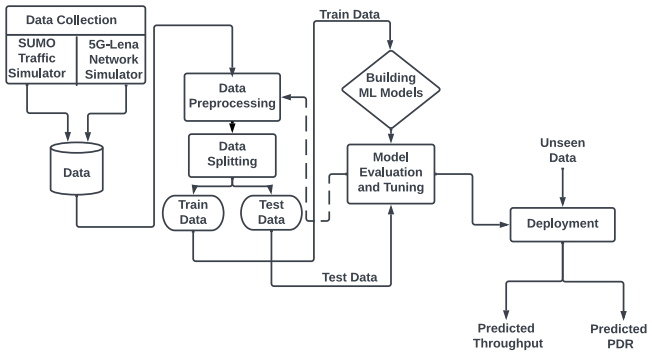


Figure 1: A flowchart of the proposed methodology

II. System Model

We consider NR-V2X Uu interface [2], where vehicles communicate with a base station. We simulate communications between one vehicle and one base station through SUMO traffic simulator and 5G-Lena network simulator. SUMO provides realistic mobility, whereas 5G-Lena ensures communications between the UE and the base station. SUMO is an open-source, microscopic, multi-modal traffic simulator used to generate vehicle trajectories [10]. 5G-Lena is an E2E open-source NR system-level simulator [11].

Simulation is used to create the dataset for the training of the prediction models. The data extracted from the simulation includes MCS, distance-to-base station, SINR, and User Datagram Protocol (UDP) packet size. Units of distance-to-base station, SINR, and UDP packet size are in meters(m), dB, and bytes, respectively. Moreover, MCS is adjusted to obtain optimal network performance. Distance-to-base station is calculated by using coordinates of the vehicle obtained from SUMO and coordinates of the base station obtained from the network simulator. SINR value is obtained at the end of simulations. To increase variety of data, different MCS and packet size are set in simulations and are added to dataset without any processing.

III. Machine Learning Based Algorithm

In this section, we provide our ML based NR-V2X network throughput and PDR prediction algorithm by including the details about data preparation and preprocessing, as well as hyperparameter selection for best performance. The algorithm is based on the iterative comparison of various ML algorithms. These ML algorithms include SVR, DNN, RF, GBM, LGBM, XGBoost and CatBoost. Furthermore, we select MCS, distance-to-base station, SINR, and packet size as inputs for the models, and employ root-mean-square error (RMSE), R-squared (R2 score) and mean absolute error (MAE) as metrics for comparing the performances of the models.

Figure 1 shows the flowchart of the proposed methodology. First, simulation scenarios are created for NR-V2X through SUMO traffic simulator and 5G-Lena network simulator. Raw data are collected for SINR, PDR, latency, MCS, throughput, packet size and distance-to-base station over four different scenarios. Then, the data is prepared for ML models with the step of data preprocessing. The third step is to divide the data into training and test data. After splitting our data, ML models for regression are built and executed for predicting throughput and PDR via offline training. Moreover, the regression models are assessed with RMSE, R2 score and MAE performance metrics and tuned iteratively to obtain the best prediction performance. Finally, the models are tested by using test data and prepared for deployment.

A. Dataset

Our objective is to train our prediction models with the simulation generated MCS, distance-to-base station, SINR and packet size data inputs. Hence, we gather data from diverse scenarios over four different simulation areas, three in Istanbul and one in Berlin, created by using SUMO and 5G-Lena simulators. One Vehicle User Equipment (VUE) emulating a single vehicle and one base station are used in the simulations. Moreover, the dataset is diversified by adjusting different MCS, base station location, vehicle location, Tx power and packet size for scenarios. Our dataset is available on GitHub[1].

B. Data Preprocessing

The data is analyzed and prepared for training purposes. First, duplicate data is removed, and thus the total number of data samples in the dataset is 8716. Data imputation is applied for missing values in this dataset. In addition, we analyze the data with Pearson Correlation heatmap where the correlation values between features are observed to be below 70%, which indicates the diversity of the inputs. We use the min-max scalar as a feature scaling method for DNN and SVR algorithms to normalize feature values to fit within the [0,1] range since the features inherit different units. The dataset is then split into 80% training data and 20% test data. On the other hand, we investigate feature importance ranking with RF algorithm for network throughput prediction and PDR prediction.

C. Machine Learning Algorithms

Considered ML models used for throughput prediction and PDR prediction are detailed below. We use Python libraries of Keras and Scikit-Learn to train, test, and validate models.

1) SVR: It aims to find the best decision boundary to separate the data into different classes, as a type of Support

[1]https://github.com/aslirey/ML-Aided-NR-V2X-QoS-Predictions

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Vector Machine (SVM) [12]. Kernel parameter is used for finding a hyperplane in the higher dimensional space and is set to radial basis function (rbf) for predicting throughput and PDR.

2) DNN: It is a type of artificial neural network which consists of an input layer, an output layer, and hidden layers. Each layer includes interconnected nodes, called neurons, where each node receives input from the prior layer and generates output for the next layer. Hidden layers perform computations with the aid of weight and bias parameters on the data to extract features and make predictions [13]. Number of hidden layers, activation function, optimizer, batch size, and epoch size of DNN for throughput prediction and PDR prediction are set to 3, 'relu', 'adam', 10 and 150, respectively. For predicting throughput, number of hidden units in the first layer is 16 and in the other two layers is 32, whereas for predicting PDR, number of hidden units in the first layer is 32 and in the other two layers is 64.
3) RF: It consists of many decision trees and a bagging technique. Therefore, decision trees in the forest are built independently and in parallel [14]. Number of trees in the forest and min samples split of RF parameters for predicting throughput and PDR are 100 and 2, respectively.
4) GBM: It includes a sequence of weak models such as decision trees, where each model learns from the errors of the previous model. The final prediction is made by taking account into all predictions from the models in the ensemble [15]. Number of boosting stages for predicting throughput and PDR is set to 300 and 250, respectively whereas learning rate and max depth are set to 0.3 and 4 for both forecasting cases, respectively.
5) XGBoost: It is an optimized gradient boosting method through parallelization, tree pruning, and regularization and also uses a technique called similarity score to determine root and nodes [16]. Booster, learning rate, and max depth of XGBoost parameters for predicting throughput and PDR are adjusted to 'gbtree', 0.3 and 4, respectively. Number of gradient-boosted trees for predicting throughput and PDR is set to 100 and 300.
6) CatBoost: It is an optimized gradient boosting method which is able to overcome categorical data, overfitting, and missing values [17]. Depth, learning rate, and number of iterations of CatBoost parameters for predicting throughput and PDR are set to 6, 0.3, and 200, respectively.
7) LGBM: It is an optimized version of GBM and utilizes the leaf-wise tree growth method to enhance the performance. It is faster than GBM since it uses histogram-based optimization [18]. Number of iterations, boosting type, learning rate, and max depth of LGBM parameters in order to forecast throughput and PDR are 200, 'gbdt', 0.3, and 6, respectively.

ML algorithms are trained with 80% offline training data for throughput and PDR prediction. The R2 score, RMSE, and MAE are utilized to evaluate the performance of the trained regression models and the hyper-parameters of the models are iteratively adjusted to achieve the best prediction results, by applying exhaustive search technique. Finally, the prediction accuracy of the models is tested with 20% test data.

IV. Numerical Results and Performance Evaluation

We present feature importance of features in our dataset, and assess the success of ML algorithms in terms of R2 score, RMSE, and MAE.

Figure 2 illustrates the weight of the features in the dataset for predicting throughput and PDR by using the RF algorithm. SINR has the highest importance value of 0.67 for predicting PDR, whereas packet size has the lowest effect on PDR prediction, with a value of 0.03. For forecasting throughput, packet size and SINR are the two most significant features with values of 0.43 and 0.39, respectively. On the other hand, the distance is found to be the weakest feature, with a value of 0.06.

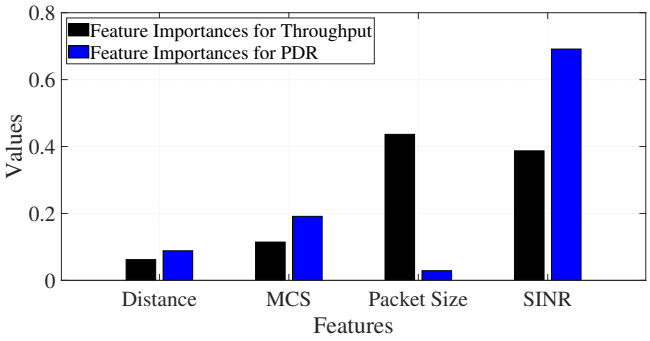
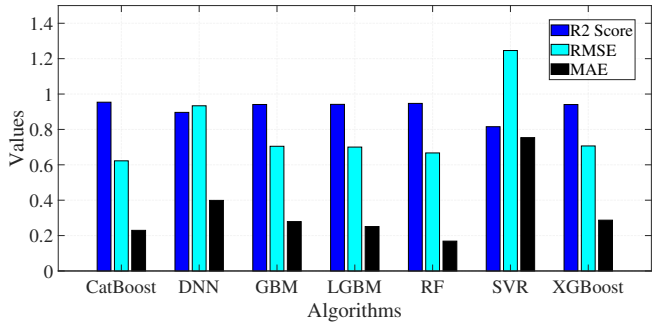


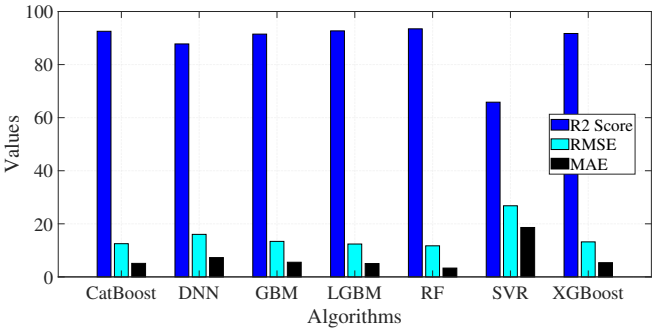
Figure 2: Importance Ranking for Predictions

Figure 3 shows values of R2 score, RMSE, and MAE of the algorithms utilized for predicting throughput and PDR. These values show how well the model fits a dataset for regression analysis. MAE handles all errors equally, whereas RMSE is more sensitive to outliers and can punish large errors. Moreover, R2 score represents the percentage of the variation that the model can explain in the dependent variable. A model with a higher R2 score, lower RMSE and MAE values demonstrates better compliance with the data, yielding better predictions performance. Units of RMSE and MAE are Mbps for throughput and percentage for PDR. In Figure 3(a), the R2 scores of all ensemble learning models for predicting throughput are over 90%. In addition, the R2 score of the CatBoost is the highest score at 95.3% for forecasting throughput, whereas R2 score of the DNN is 90.8%. On the other hand, the SVR model has the lowest R2 score for throughput with 81.5%. The RMSE values of all ensemble learning models for throughput prediction are below 0.71 and the CatBoost has the lowest value at 0.62. The SVR algorithm has the highest

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(a) R2 score, RMSE and MAE values for predicting throughput



(b) R2 score, RMSE and MAE values for predicting PDR

Figure 3: Performances of Machine Learning Algorithms

RMSE value for predicting throughput, compared to the other models with a value of 1.25. The performance of the DNN model is the second worst as regards the RMSE value for throughput prediction. The first two lowest MAE values belong to RF and the CatBoost for predicting throughput, with values of 0.17 and 0.23, respectively. It is seen that the SVR has the worst performance for forecasting throughput, with a value of 0.76 taking into account to MAE values of the models. In Figure 3(b), all ensemble learning models are demonstrated to achieve R2 scores of over 90% for predicting PDR. RF has the highest score at 93.4% to estimate PDR. R2 score of DNN is 85.3%. The SVR model has the lowest R2 score for PDR prediction with 64.3%, but all ensemble learning models have the RMSE values below 14 for PDR prediction. On the other hand, among the ensemble learning models, RF has the lowest RMSE value at 11.7. The SVR algorithm has a higher RMSE value for predicting PDR compared to the other models, with a value of 25.8. Moreover, RF and the LGBM have the lowest MAE values for predicting PDR, with values of 3.32 and 5, respectively. The SVR has the worst performance for predicting PDR, with a MAE value of 18.62.

V. Conclusions

In this paper, we propose a ML based throughput and PDR prediction scheme, which aims to provide predictive QoS information to adopt V2X application parameters respectively. We utilize custom generated simulation data consists of NR-V2X MCS, SINR, packet size and distance-to-base station to train and validate models. We rank the feature importance to predict the QoS and show that CatBoost and RF are the best algorithms to predict throughput, and PDR respectively. Our results depict that packet size and SINR are the most crucial features for predicting throughput and PDR, respectively. Future work will investigate the prediction of other V2X network QoS parameters to enable reliable decision making regarding application reliability for given network conditions.

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